This assignment will not be graded and is only for practice.

Level 1

1) Choose the set of correct options. ***1 point***

- [ ] If S is a spanning set of the vector space V, then $S \cup \{v\}$ must be a spanning set of V, for all $v \in V$.
- [ ] Span of an empty set is the zero vector space.
- [ ] If S is a spanning set of the vector space V, then $S \setminus \{v\}$ must be a spanning set of V, for all $v \in S$.
- [ ] If S is a spanning set of the vector space V, then $S \cup \{v\}$ may not be a spanning set of V, for all $v \in V$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

If S is a spanning set of the vector space V, then $S \cup \{v\}$ must be a spanning set of V, for all $v \in V$.
Span of an empty set is the zero vector space.

2) Let S be the set of matrices where $S = \{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\}$. Consider the following statements:

Statement P: $Span(S)$ is the vector space consisting of only lower triangular square matrices of order 2.
Statement Q: $Span(S)$ is the vector space consisting of only upper triangular square matrices of order 2.
Statement R: $Span(S)$ is the vector space consisting of all the square matrices of order 2.
Statement S: $Span(S)$ is the vector space consisting of only scalar matrices of order 2.

Find the number of correct statements.

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

3) If S is a basis of R^2, then what is the cardinality of S?

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

4) Let $S = \{(1,2),(1,3)\}$ and $span(S) = \mathrm{R}^n$. Then find the value of n.

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

5) Let S be the set of matrices where $S = \{\begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}, \begin{bmatrix}0 & 0\\0 & 1\end{bmatrix}\}$. Consider the following statements:

Statements P: $Span(S)$ is the vector space consisting of only the Identity matrix of order 2.
Statements Q: $Span(S)$ is the vector space consisting of only diagonal matrices of order 2.
Statements R: $Span(S)$ is the vector space consisting of all the square matrices of order 2.
Statements S: $Span(S)$ is the vector space consisting of only scalar matrices of order 2.

Find the number of correct statements.

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

6)

1 point

Consider the following set of vectors $S = \{(1,1,-1),(-1,1,1),(0,\frac{1}{2},0),(0,1,-2),(1,0,-2)\}$. Choose the set of correct options.

- [ ] The set { (1, 1, -1), (-1, 1, 1), (0, $\frac{1}{2}$,0)} is not a basis of R^3.
- [ ] The set {(0, $\frac{1}{2}$,0), (0, 1, -2), (1, 0 ,-2),(1, 0, -2)} is a basis of R^3.
- [ ] The set {(-1, 1, 1), (0, $\frac{1}{2}$,0), (0, 1, -2)} is a basis of R^3.
- [ ] None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The set { (1, 1, -1), (-1, 1, 1), (0, $\frac{1}{2}$,0)} is not a basis of R^3.
The set {(-1, 1, 1), (0, $\frac{1}{2}$,0), (0, 1, -2)} is a basis of R^3.

7) Consider a set of vectors $S = \{(1,2),(-1,2),(3,1),(-3,1)\}$ in R^2. Choose the set of correct options. ***1 point***

- [ ] The set { (1, 2) ,(-1, 2), (3, 1)} is a basis of R^2.
- [ ] The set { (-1, 2), (3, 1) } is a basis of R^2.
- [ ] The set { (3, 1), (-3, 1) } is a basis of R^2.
- [ ] None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The set { (-1, 2), (3, 1) } is a basis of R^2.
The set { (3, 1), (-3, 1) } is a basis of R^2.

Level 2

8) Choose the set of correct options. ***1 point***

- [ ] If a subset S of R^2 contains only two elements then S is a basis of R^2.
- [ ] If a subset S of R^3 contains only one elements then S can never be a basis of R^3.
- [ ] If a subset S of R contains only one non zero element then S is a basis of R.
- [ ] If S_1 and S_2 are two bases of R^3 then $S_1 = S_2$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

If a subset S of R^3 contains only one elements then S can never be a basis of R^3.
If a subset S of R contains only one non zero element then S is a basis of R.

9) Let V be the subspace of R^3 defined as follows: ***1 point***

$$V = \{(x, y, z) \mid x = y - z, \text{ and } x, y, z \in \mathrm{R}\}$$

Choose the set of correct options from the following.

- [ ] $\{(1,1,0),(1,0,-1)\}$ is a linearly independent set of V.
- [ ] $\{(1,1,0),(1,0,-1),(0,1,1)\}$ is a linearly independent set of V.
- [ ] $\{(0,1,1),(1,0,-1)\}$ is a spanning set of V.
- [ ] $\{(1,1,0)\}$ is a spanning set of V.

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\{(1,1,0),(1,0,-1)\}$ is a linearly independent set of V.
$\{(0,1,1),(1,0,-1)\}$ is a spanning set of V.

10) Let V and W be vector spaces which are defined as follows: ***1 point***

$V = \{(x, y) \mid y = mx, \text{ where } m = 0 \text{ and } x, y, m \in \mathbb{R}\}$ with usual addition and scalar multiplication as in $\mathbb{R}^2$. $W = \{(x, y) \mid x = 0\}$ with usual addition and scalar multiplication as in $\mathbb{R}^2$.

Choose the correct set of options.

- [ ] The set $\{(1, m), (\frac{1}{m}, 1)\}$ is a linearly independent set in V.
- [ ] The set $\{(1, m), (\frac{1}{m}, 1)\}$ is a spanning set for V.
- [ ] The set $\{(1, m)\}$ is a linearly independent set in V.
- [ ] The set $\{(\frac{1}{m}, 1)\}$ is a linearly independent set in V.
- [ ] The set $\{(0, 1), (0, 2)\}$ is a linearly independent set in W
- [ ] The set $\{(0, 1)\}$ is a linearly independent set in W.
- [ ] The set $\{(0, 5)\}$ is a spanning set for W.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The set $\{(1, m), (\frac{1}{m}, 1)\}$ is a spanning set for V.
The set $\{(1, m)\}$ is a linearly independent set in V.
The set $\{(\frac{1}{m}, 1)\}$ is a linearly independent set in V.
The set $\{(0, 1)\}$ is a linearly independent set in W.
The set $\{(0, 5)\}$ is a spanning set for W.

Your score is: 0/10